

Algebraic Independence in the Affine Case of Erdős Problem 270

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Erdős Problem 270 asks whether the series of reciprocals of $(n+1)(n+2)\cdots(n+f(n))$ must be irrational whenever $f(n) \rightarrow \infty$. Crmarić and Kovač showed that the answer is negative without a regularity assumption, while the problem remains open for nondecreasing f . Erdős and Graham singled out the simplest such choice, $f(n) = n$. We study the entire positive affine subfamily $f(n) = an + b$, whose members are

$$C_{a,b} := \sum_{n=1}^{\infty} \frac{n!}{((a+1)n+b)!}$$

for $a \geq 1$ and $b \geq 1 - a$. For each fixed a , we prove that $C_{a,0}, \dots, C_{a,a-1}, C_{a,a+1}$ are algebraically independent over $\mathbb{Q}$ and that every other admissible $C_{a,b}$ is a rational affine combination of them. Thus the fixed-slope family has transcendence degree $a+1$; in particular, every positive affine case of Problem 270 is transcendental. We begin with an elementary irrationality proof for the Erdős–Graham example and its extension to $0 \leq b \leq a$. A Gaussian integral representation then gives a short transcendence proof for the original example. For the general theorem, we pass to hypergeometric E -functions and combine algebraic-independence results of Salikhov and Salikhov–Viskina with a logarithmic obstruction at infinity and Beukers’s specialization theorem.

1 Irrationality of the Erdős–Graham example

Erdős and Graham ended their chapter on irrationality and transcendence with the question now known as Erdős Problem 270. Given a positive integer-valued function f with $f(n) \rightarrow \infty$, they considered

$$S(f) := \sum_{n=1}^{\infty} \frac{1}{(n+1)(n+2)\cdots(n+f(n))},$$

and asked whether $S(f)$ must be irrational. They expected an affirmative answer when f is nondecreasing [1, p. 66]. Crmarić and Kovač later showed that some regularity assumption is necessary: as f ranges over all such functions, $S(f)$ takes every value in $(0, \infty)$ [3, Theorem 1]. The nondecreasing problem remains open [2].

Erdős and Graham singled out the simplest nondecreasing choice, $f(n) = n$. They described the resulting series as difficult to treat and wrote that such series were surely irrational [1, pp. 65–66]. Crmarić and Kovač later returned to this example and reported that they knew no proof of its irrationality [3, p. 55, following (1.1)]. After their paper appeared, Crmarić and Kovač found a short proof, which Kovač posted on the Erdős Problems forum in July 2026 [4]. We obtained the argument below independently, before learning of their post. The series is the first member of the affine family considered here:

$$S(f) = C := \sum_{n=1}^{\infty} \frac{n!}{(2n)!} = \sum_{n=1}^{\infty} \frac{1}{D_n}, \quad D_n := \frac{(2n)!}{n!}.$$

Then

$$\frac{D_{n+1}}{D_n} = \frac{(2n+2)(2n+1)}{n+1} = 4n+2,$$

so $D_n \mid D_{n+1}$. If

$$P_N := \sum_{n=1}^N \frac{1}{D_n}, \quad R_N := C - P_N,$$

then $D_N P_N \in \mathbb{Z}$, while

$$\begin{aligned} 0 < D_N R_N &= \sum_{k=1}^{\infty} \frac{D_N}{D_{N+k}} \\ &= \sum_{k=1}^{\infty} \frac{1}{(4N+2)(4N+6) \cdots (4N+4k-2)} \\ &\leq \sum_{k=1}^{\infty} \frac{1}{(4N+2)^k} = \frac{1}{4N+1}. \end{aligned}$$

Thus $D_N R_N \rightarrow 0$.

Suppose that $C = p/q$, where $p \in \mathbb{Z}$ and $q \in \mathbb{Z}_{>0}$. Then

$$D_N R_N = \frac{p D_N}{q} - D_N P_N$$

is positive and belongs to $q^{-1}\mathbb{Z}$. It must therefore be at least $1/q$, contradicting $D_N R_N \rightarrow 0$. Hence C is irrational.

2 Irrationality for $0 \leq b \leq a$

For integers $a \geq 1$ and $b \geq 0$, the affine choice $f(n) = an + b$ gives

$$S(f) = \sum_{n=1}^{\infty} \frac{1}{(n+1) \cdots ((a+1)n+b)} = \sum_{n=1}^{\infty} \frac{n!}{((a+1)n+b)!} =: C_{a,b}.$$

These are the affine cases of Problem 270 with nonnegative intercept. The preceding argument extends to the range $0 \leq b \leq a$. Fix a and b in this range, and put

$$D_n := \frac{((a+1)n+b)!}{n!}.$$

Then

$$\frac{D_{n+1}}{D_n} = \frac{\prod_{j=1}^{a+1} ((a+1)n+b+j)}{n+1}.$$

The factor with $j = a + 1 - b$ is $(a + 1)(n + 1)$. Cancelling $n + 1$ gives the positive integer

$$r_n := \frac{D_{n+1}}{D_n} = (a + 1) \prod_{\substack{1 \leq j \leq a+1 \\ j \neq a+1-b}} ((a + 1)n + b + j).$$

Thus $D_n \mid D_{n+1}$. Moreover, $r_n \geq a + 1 \geq 2$, and every factor in the product defining r_n increases with n . Hence r_n is increasing and tends to infinity. Let

$$P_N := \sum_{n=1}^N \frac{1}{D_n}, \quad R_N := C_{a,b} - P_N.$$

Since $D_1 \mid D_2 \mid \cdots \mid D_N$, we have $D_N P_N \in \mathbb{Z}$. Moreover,

$$\begin{aligned} 0 < D_N R_N &= \sum_{k=1}^{\infty} \frac{1}{r_N r_{N+1} \cdots r_{N+k-1}} \\ &\leq \sum_{k=1}^{\infty} \frac{1}{r_N^k} = \frac{1}{r_N - 1} \rightarrow 0. \end{aligned}$$

If $C_{a,b} = p/q$, then $D_N R_N$ is positive and belongs to $q^{-1}\mathbb{Z}$, so it is at least $1/q$. This is impossible for large N . Hence $C_{a,b}$ is irrational for every $a \geq 1$ and $0 \leq b \leq a$.

3 Transcendence of $C_{1,0}$ and $C_{1,1}$

The elementary proof above establishes the irrationality of $C_{1,0}$. Its Gaussian integral representation leads to an independent proof of transcendence. Crmarić and Kovač record the identity [3, p. 55 and footnote 1]

$$C_{1,0} = e^{1/4} \int_0^{1/2} e^{-t^2} dt.$$

They obtain it by expanding $e^{1/4-t^2}$ and integrating termwise.

In a 2023 MathOverflow discussion of the neighboring integral $\int_0^1 e^{-t^2} dt$, Weber and Pinelis [6] considered the E -function

$$y(x) := \frac{1}{\sqrt{x}} \int_0^{\sqrt{x}} e^{-t^2} dt = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} \frac{x^n}{n!}$$

and used the Siegel–Shidlovsky theorem to prove that $y(1)$ and e^{-1} are algebraically independent. The differential system in their argument is ordinary at every nonzero algebraic point, so applying the same theorem at $x = 1/4$ gives the algebraic independence of $y(1/4)$ and $e^{-1/4}$. Since

$$y(1/4) = 2 \int_0^{1/2} e^{-t^2} dt, \quad C_{1,0} = \frac{y(1/4)}{2e^{-1/4}},$$

an algebraic value of $C_{1,0}$ would give a nontrivial algebraic relation between $y(1/4)$ and $e^{-1/4}$. Hence $C_{1,0}$ is transcendental. The MathOverflow discussion did not mention either the factorial series or this specialization.

To pass from this special argument to the full family, we express the case $a = 1$ as a hypergeometric E -function. This introduces the method used in Section 4 and treats $C_{1,1}$ as well. For $b \in \{0, 1\}$, define

$$F_{1,b}(z) := \sum_{n=0}^{\infty} \frac{n!}{(2n+b)!} z^n.$$

Then

$$F_{1,b}(1) = 1 + C_{1,b}.$$

Writing $\beta = b + 1/2$, direct coefficient comparison gives

$$F_{1,b}(z) = {}_1F_1\left(1; \beta; \frac{z}{4}\right).$$

This is an E -function. If $h(x) = {}_1F_1(1; \beta; x)$, Kummer's equation reads

$$xh'' + (\beta - x)h' - h = 0.$$

The left side is the derivative of $xh' + (\beta - 1 - x)h$. Evaluation at $x = 0$ therefore gives

$$xh'(x) + (\beta - 1 - x)h(x) = \beta - 1.$$

After substituting $x = z/4$,

$$zF'_{1,b}(z) + \left(\beta - 1 - \frac{z}{4}\right) F_{1,b}(z) = \beta - 1. \quad (1)$$

Thus $(1, F_{1,b})^\top$ satisfies a first-order system over $\overline{\mathbb{Q}}(z)$ whose only finite singularity is 0.

The functions 1 and $F_{1,b}$ are linearly independent over $\overline{\mathbb{Q}}(z)$. Otherwise $F_{1,b}$ would be rational; because it is entire, it would be a polynomial, contrary to its infinitely many nonzero Taylor coefficients. Beukers [7, Corollary 1.4] proves that E -functions which are linearly independent over $\overline{\mathbb{Q}}(z)$ remain linearly independent after evaluation at a nonzero algebraic point that is ordinary for their differential system. Applied at $z = 1$, this gives linear independence of 1 and $F_{1,b}(1)$ over $\overline{\mathbb{Q}}$. Hence $F_{1,b}(1)$ is transcendental, and so are both $C_{1,0}$ and $C_{1,1}$. The case $b = 0$ recovers the preceding conclusion, while $b = 1$ gives the other case in the range $0 \leq b \leq a$ when $a = 1$.

4 The affine family

Theorem 1. *For every integer $a \geq 1$, the numbers*

$$C_{a,0}, C_{a,1}, \dots, C_{a,a-1}, C_{a,a+1}$$

are algebraically independent over $\overline{\mathbb{Q}}$. Every $C_{a,b}$ with $b \geq 1 - a$ is a rational affine combination of these $a + 1$ numbers, and

$$\text{trdeg}_{\overline{\mathbb{Q}}} \overline{\mathbb{Q}}(C_{a,b} : b \geq 1 - a) = a + 1.$$

Moreover, every $C_{a,b}$ in this range is transcendental.

We first prove the transcendence assertion for every admissible intercept. The same hypergeometric framework will then be used to determine all algebraic relations at each fixed slope.

For all integers $a \geq 1$ and $b \geq 0$, define

$$F_{a,b}(z) := b! \sum_{n=0}^{\infty} \frac{n!}{((a+1)n+b)!} z^{an}. \quad (2)$$

Then

$$F_{a,b}(1) = 1 + b!C_{a,b}. \quad (3)$$

For $b \geq 0$, it is therefore enough to prove that $F_{a,b}(1)$ is transcendental. Beukers's theorem reduces this to finding a differential system, ordinary at 1, whose components include $F_{a,b}$ and are linearly independent over $\overline{\mathbb{Q}}(z)$.

4.1 The range $0 \leq r \leq a$

Assume first that $0 \leq r \leq a$. Put

$$J_{a,r} := \{1, \dots, a+1\} \setminus \{a+1-r\}, \quad \beta_j := \frac{r+j}{a+1} \quad (j \in J_{a,r}).$$

The omitted parameter is the unique one equal to 1. The multiplication formula gives

$$F_{a,r}(z) = {}_1F_a \left(\begin{matrix} 1 \\ (\beta_j)_{j \in J_{a,r}} \end{matrix}; \frac{z^a}{(a+1)^{a+1}} \right). \quad (4)$$

Indeed,

$$\prod_{j=1}^{a+1} \left(\frac{r+j}{a+1} \right)_n = \frac{((a+1)n+r)!}{r!(a+1)^{(a+1)n}},$$

and deleting the factor $(1)_n = n!$ yields (4).

To match the notation of the cited results, set

$$\rho_a := \frac{a}{(a+1)^{(a+1)/a}}, \quad \lambda_j := \beta_j - 1,$$

and define

$$\varphi(w) := \sum_{n=0}^{\infty} \frac{(w/a)^{an}}{\prod_{j \in J_{a,r}} (\lambda_j + 1)_n}.$$

Then $F_{a,r}(z) = \varphi(\rho_a z)$. This is the $l = 0$, $t = a$ case of Salikhov's formula [8, eq. (1)]. Here $l = 0$ means that Salikhov's numerator contains no Pochhammer factors; in standard hypergeometric notation the parameter 1 in (4) remains because $(1)_n/n! = 1$.

Let $\theta = z d/dz$ and define

$$W_{a,r} := \text{span}_{\mathbb{C}(z)} \{1, F_{a,r}, \theta F_{a,r}, \dots, \theta^{a-1} F_{a,r}\}. \quad (5)$$

The key functional statement is

$$\dim_{\mathbb{C}(z)} W_{a,r} = a + 1. \quad (6)$$

For odd a , Salikhov's Theorem 1 applies to this $l = 0$ function [8, Theorem 1]. The parameters are rational, and none is a negative integer. The remaining hypothesis requires that their multiset modulo integers not be invariant under translation by $1/d$ for any divisor $d > 1$ of a . In the present notation, this multiset is

$$S_a = \left\{ \frac{1}{a+1}, \dots, \frac{a}{a+1} \right\} \subset \mathbb{Q}/\mathbb{Z}.$$

If $d > 1$ divides a , then $\gcd(d, a+1) = 1$, so $1/d$ does not lie in the subgroup $(a+1)^{-1}\mathbb{Z}/\mathbb{Z}$ containing S_a . Translation by $1/d$ therefore cannot preserve S_a , which verifies Salikhov's parameter condition. The theorem gives algebraic independence of $\varphi(\alpha), \varphi'(\alpha), \dots, \varphi^{(a-1)}(\alpha)$ for every nonzero algebraic α .

When a is even, the theorem of Salikhov–Viskina applies with $t = a$. If the λ_j are rational and not negative integers, it states that for every nonzero algebraic α the numbers $1, \varphi(\alpha), \delta\varphi(\alpha), \dots, \delta^{a-1}\varphi(\alpha)$ are linearly independent over $\overline{\mathbb{Q}}$, where $\delta = w d/dw$ is their Euler operator [9, p. 833]. Unlike Salikhov's odd-order result, this gives linear rather than algebraic independence; it includes the constant 1 and imposes no translation condition.

In either parity, the cited theorem implies that the values of $1, F_{a,r}, \theta F_{a,r}, \dots, \theta^{a-1} F_{a,r}$ are linearly independent over $\overline{\mathbb{Q}}$ at every nonzero algebraic point. At such a point α , the Euler derivatives $(\theta^k F_{a,r})(\alpha)$ and the ordinary derivatives $F_{a,r}^{(k)}(\alpha)$ are related by a triangular algebraic matrix with diagonal entries $1, \alpha, \dots, \alpha^{a-1}$. The two forms of the independence statement are therefore equivalent.

Descent lemma. *Let $g_1, \dots, g_m \in \overline{\mathbb{Q}}[[z]]$. If these series are linearly dependent over $\mathbb{C}(z)$, then they are linearly dependent over $\overline{\mathbb{Q}}(z)$.*

Proof. Clear the denominators in a nonzero relation and let D be the largest degree of its polynomial coefficients. Writing those coefficients as $\sum_{d=0}^D c_{i,d} z^d$ and comparing Taylor coefficients gives a homogeneous linear system over $\overline{\mathbb{Q}}$ in the finitely many unknowns $c_{i,d}$. Its row space has a finite basis, so the system has the same kernel as a finite matrix over $\overline{\mathbb{Q}}$. If its kernel over $\mathbb{C}$ is nonzero, then its kernel over $\overline{\mathbb{Q}}$ is nonzero, and the corresponding polynomial coefficients give the required relation. $\square$

Every function in (5) has algebraic Taylor coefficients, so the descent lemma implies (6). Otherwise a relation over $\mathbb{C}(z)$ would descend to $\overline{\mathbb{Q}}(z)$. Specializing at a nonzero algebraic point where its coefficients are defined and not all zero would contradict the cited value independence.

Applying the value-independence statement at $z = 1$ proves that $F_{a,r}(1)$, and hence $C_{a,r}$ by (3), is transcendental for $0 \leq r \leq a$.

The same statement handles affine functions with negative intercept that remain positive for every $n \geq 1$. Suppose that $1 - a \leq b < 0$ and put $r = a + 1 + b$, so that $a \geq 2$ and $2 \leq r \leq a$.

Shifting the summation index by writing $n = m + 1$ gives

$$\begin{aligned} C_{a,b} &= \sum_{m=0}^{\infty} \frac{(m+1)!}{((a+1)m+r)!} \\ &= \frac{1}{r!} \left(F_{a,r}(1) + \frac{1}{a} \theta F_{a,r}(1) \right). \end{aligned}$$

If $C_{a,b}$ were algebraic, this identity would give a nontrivial linear relation over $\overline{\mathbb{Q}}$ among 1 , $F_{a,r}(1)$, and $\theta F_{a,r}(1)$. Hence $C_{a,b}$ is transcendental for $1 - a \leq b < 0$ as well.

To treat $b > a$, we compare the formal solutions of the relevant differential equations at infinity. The generalized hypergeometric equation for (4) can be written as

$$\left[\prod_{j \in J_{a,r}} \left(\frac{\theta}{a} + \beta_j - 1 \right) - \frac{z^a}{(a+1)^{a+1}} \right] F_{a,r} = \prod_{j \in J_{a,r}} (\beta_j - 1). \quad (7)$$

The right side is nonzero because the parameter equal to 1 was removed. To describe the formal system at infinity, put

$$x := z^{-1}, \quad K := \mathbb{C}((x)), \quad \kappa_a := (a+1)^{-(a+1)},$$

so that $\theta = -x d/dx$ on K , and write

$$c_{a,r} := \prod_{j \in J_{a,r}} (\beta_j - 1), \quad P_{a,r}(u) := \prod_{j \in J_{a,r}} (u + \beta_j - 1) = \sum_{j=0}^a p_j u^j, \quad p_a = 1.$$

Let

$$Y_0 := 1, \quad Y_j := \theta^{j-1} F_{a,r} \quad (1 \leq j \leq a).$$

Equation (7) is equivalent to the augmented companion system

$$\theta Y_0 = 0, \quad \theta Y_j = Y_{j+1} \quad (1 \leq j < a),$$

$$\theta Y_a = a^a c_{a,r} Y_0 + a^a (\kappa_a z^a - p_0) Y_1 - \sum_{j=1}^{a-1} a^{a-j} p_j Y_{j+1}.$$

After expanding rational functions at infinity, this system defines a rank- $(a+1)$ differential module $\mathcal{M}_{a,r}$ over K .

Lemma 4.1 (Zero-exponential formal component). *The Levelt–Turrittin decomposition of $\mathcal{M}_{a,r}$ is*

$$\mathcal{M}_{a,r} = \mathcal{M}_0 \oplus \bigoplus_{\zeta^a=1} \mathcal{M}_{\zeta},$$

where $\mathcal{M}_0$ has rank one and exponential part 0, while each $\mathcal{M}_{\zeta}$ has rank one and exponential part $\rho_a \zeta z$. The zero summand has the formal solution vector

$$\widehat{Y}^{(0)} := \left(1, \widehat{F}_{a,r}^{(0)}, \theta \widehat{F}_{a,r}^{(0)}, \dots, \theta^{a-1} \widehat{F}_{a,r}^{(0)} \right)^{\top},$$

where

$$\widehat{F}_{a,r}^{(0)}(z) \in z^{-a}\mathbb{C}[[z^{-a}]]. \quad (8)$$

More explicitly, if

$$G = r_0 + \sum_{k=0}^{a-1} r_{k+1} \theta^k F_{a,r} \in W_{a,r}, \quad r_0, \dots, r_a \in \mathbb{C}(z),$$

then its zero-exponential projection is

$$\pi_0(G) = r_0 + \sum_{k=0}^{a-1} r_{k+1} \theta^k \widehat{F}_{a,r}^{(0)} \in K.$$

The projection extends K -linearly to $K \otimes_{\mathbb{C}(z)} W_{a,r}$ and commutes with θ . In particular, $\pi_0(G)$ is a Laurent series without powers of $\log z$.

Proof. Writing $\widehat{F}_{a,r}^{(0)} = \sum_{m \geq 1} h_m z^{-am}$ in (7) gives

$$h_1 = -\frac{c_{a,r}}{\kappa_a}, \quad h_{m+1} = \frac{P_{a,r}(-m)}{\kappa_a} h_m \quad (m \geq 1). \quad (9)$$

This recurrence constructs the solution (8). It is unique among Laurent-series particular solutions. Indeed, the difference of two such solutions would be a nonzero Laurent-series solution of the homogeneous equation. If dz^ν were its leading monomial, the term $-\kappa_a z^a dz^\nu$ could not be cancelled by $P_{a,r}(\theta/a)$, which does not raise exponents. Consequently $\widehat{Y}^{(0)}$ is a nonzero formal solution of the augmented system with exponential part 0.

The homogeneous equation associated with (7) has characteristic equation

$$\lambda^a = \frac{a^a}{(a+1)^{a+1}},$$

so its a characteristic numbers are the distinct nonzero numbers $\rho_a \zeta$. Its formal homogeneous solutions have the form

$$e^{\rho_a \zeta z} z^{\mu_\zeta} \mathbb{C}[[z^{-1}]], \quad \zeta^a = 1,$$

and are log-free because every characteristic number is simple; see Salikhov [8, Remark 2, p. 206]. Each of the a distinct nonzero exponential parts therefore contributes a rank-one summand.

The Levelt–Turrittin theorem decomposes $\mathcal{M}_{a,r}$ as a direct sum of differential modules indexed by their exponential parts [10, Chapter 3, Theorem 3.1]. No ramification is needed because 0 and the polynomials $\rho_a \zeta z$ lie over K . The nonzero exponential summands account for rank a , and $\widehat{Y}^{(0)}$ supplies a nonzero solution in the zero-exponential summand. Since $\mathcal{M}_{a,r}$ has rank $a+1$, the zero summand has rank exactly one, and its formal solution space is spanned by $\widehat{Y}^{(0)}$.

By (6), the displayed expression for G in the statement is unique. Expanding the coefficients r_j in K and projecting the companion vector onto the line spanned by $\widehat{Y}^{(0)}$ gives the stated formula for $\pi_0(G)$. Because the Levelt–Turrittin summands are differential modules, this projection is K -linear and commutes with θ . Every entry of $\widehat{Y}^{(0)}$ lies in K , so the formula also shows directly that $\pi_0(G)$ contains no logarithms. $\square$

4.2 Euler reduction for $b > a$

Suppose that $b > a$ and write

$$b = q(a + 1) + r, \quad q \geq 1, \quad 0 \leq r \leq a. \quad (10)$$

Put

$$\beta_j := \frac{b + j}{a + 1}, \quad 1 \leq j \leq a + 1, \quad x := \frac{z^a}{(a + 1)^{a+1}}.$$

None of the lower parameters is equal to 1, and the multiplication formula gives

$$F_{a,b}(z) = {}_2F_{a+1} \left(\begin{matrix} 1, 1 \\ \beta_1, \dots, \beta_{a+1} \end{matrix}; x \right). \quad (11)$$

The function $F_{a,b}$ in (11) is a hypergeometric E -function: its parameters are positive rational numbers, and with two upper and $a + 1$ lower parameters, the required pullback degree is $(a + 1) - 2 + 1 = a$. Writing its differential equation in terms of $\theta_x = x d/dx$, factoring off θ_x , and evaluating at $x = 0$ to determine the resulting constant gives

$$\left[\prod_{j=1}^{a+1} \left(\frac{\theta}{a} + \beta_j - 1 \right) - \frac{z^a}{(a + 1)^{a+1}} \left(\frac{\theta}{a} + 1 \right) \right] F_{a,b} = \prod_{j=1}^{a+1} (\beta_j - 1). \quad (12)$$

This is an order- $(a + 1)$ inhomogeneous equation. Its leading ordinary-derivative coefficient is a nonzero constant multiple of z^{a+1} , so its companion system has no nonzero finite singularity. Coefficient comparison in (2) gives the Euler reduction

$$\prod_{k=1}^q \left(\frac{\theta}{a} + k \right) F_{a,b}(z) = \frac{b!}{r!} z^{-aq} \left[F_{a,r}(z) - r! \sum_{m=0}^{q-1} \frac{m!}{((a + 1)m + r)!} z^{am} \right]. \quad (13)$$

The operator on the left multiplies the coefficient of z^{an} by

$$\prod_{k=1}^q (n + k) = \frac{(n + q)!}{n!}.$$

The index shift $n \mapsto n + q$ gives the right side of (13). In particular, the differential module generated by 1 and $F_{a,b}$ contains $W_{a,r}$.

4.3 Resonance

Assume for contradiction that $F_{a,b} \in W_{a,r}$ and apply the projection π_0 of Lemma 4.1 to (13). This projection commutes with the Euler operator on the left. On the right, the contribution of $z^{-aq} F_{a,r}$ begins at $z^{-a(q+1)}$ by (8), whereas the highest term of the polynomial subtracted in (13), corresponding to $m = q - 1$, contributes the nonzero z^{-a} coefficient

$$-\frac{b!(q - 1)!}{((a + 1)(q - 1) + r)!}. \quad (14)$$

Under this assumption, the lemma shows that the zero-exponential component of $F_{a,b}$ is a Laurent series $L(z) \in \mathbb{C}((z^{-1}))$ without logarithms. The Euler operator on the left of (13) acts diagonally on monomials, and its multiplier on z^{-a} is

$$\prod_{k=1}^q (-1 + k) = 0. \quad (15)$$

It follows that the z^{-a} coefficient in the zero-exponential component of the left side of (13) must vanish, contradicting (14). Therefore

$$F_{a,b} \notin W_{a,r}. \quad (16)$$

The projected equation therefore requires a logarithmic term. Since

$$\prod_{k=1}^q \left(\frac{\theta}{a} + k \right) (z^{-a} \log z) = \frac{(q-1)!}{a} z^{-a},$$

comparison with (14) gives

$$\widehat{F}_{a,b}^{(0)}(z) = -\frac{a b!}{((a+1)(q-1)+r)!} z^{-a} \log z + \cdots, \quad (17)$$

where the omitted terms do not affect the coefficient of $z^{-a} \log z$.

Define

$$V_{a,b} := \text{span}_{\mathbb{C}(z)} \{1, F_{a,b}, \theta F_{a,b}, \dots, \theta^a F_{a,b}\}. \quad (18)$$

Equation (12) makes $V_{a,b}$ stable under θ and gives $\dim V_{a,b} \leq a+2$. Equation (13) gives $W_{a,r} \subseteq V_{a,b}$. By (6) and (16), the additional element $F_{a,b}$ is not in $W_{a,r}$, so

$$\dim V_{a,b} \geq (a+1) + 1 = a+2.$$

Hence $\dim V_{a,b} = a+2$, and

$$1, F_{a,b}, F'_{a,b}, \dots, F_{a,b}^{(a)} \quad (19)$$

are linearly independent over $\mathbb{C}(z)$. Thus (12) has minimal possible order among inhomogeneous equations over $\mathbb{C}(z)$ satisfied by $F_{a,b}$.

4.4 Values at $z = 1$

By (11), $F_{a,b}$ is an E -function, as are its derivatives. The functions in (19) are linearly independent over $\mathbb{Q}(z)$, and (12) supplies a companion system with no nonzero finite singularity. Beukers [7, Corollary 1.4] therefore gives the linear independence of their values at the ordinary algebraic point $z = 1$. In particular, $F_{a,b}(1)$ is transcendental, and (3) gives the transcendence of $C_{a,b}$.

Combining this with the cases $1-a \leq b \leq a$ treated above proves that $S(f)$ is transcendental for every positive integer-valued affine function $f(n) = an + b$ with $a \geq 1$. It remains to determine the algebraic relations among the values with the same slope.

4.5 Algebraic independence at fixed slope

Christopher D. Long suggested extending the transcendence argument to algebraic independence [5]. For a fixed slope, the entire family cannot be algebraically independent because its members satisfy affine recurrences. We will prove that the maximal independent subfamily is

$$C_{a,0}, C_{a,1}, \dots, C_{a,a-1}, C_{a,a+1} \quad (20)$$

and that

$$\text{trdeg}_{\overline{\mathbb{Q}}} \overline{\mathbb{Q}}(C_{a,b} : b \geq 1 - a) = a + 1. \quad (21)$$

Fix $a \geq 1$, continue to write $\theta = z d/dz$, and abbreviate $F_b = F_{a,b}$. Direct comparison of the coefficients in (2) gives

$$F_{b-1} = F_b + \frac{a+1}{ab} \theta F_b \quad (1 \leq b \leq a). \quad (22)$$

Indeed, the coefficient of z^{an} on the right is

$$\frac{b!n!}{((a+1)n+b)!} \left(1 + \frac{(a+1)n}{b} \right) = \frac{(b-1)!n!}{((a+1)n+b-1)!}.$$

Put $H = F_{a-1}$. Iterating (22) gives

$$F_j = P_j(\theta)H, \quad P_j(X) = \prod_{k=j+1}^{a-1} \left(1 + \frac{a+1}{ak} X \right), \quad 0 \leq j \leq a-1, \quad (23)$$

where an empty product is 1. The polynomials $P_{a-1}, P_{a-2}, \dots, P_0$ have degrees $0, 1, \dots, a-1$ and nonzero leading coefficients. Thus (23) is an invertible triangular linear transformation between

$$F_0, F_1, \dots, F_{a-1} \quad \text{and} \quad H, \theta H, \dots, \theta^{a-1} H. \quad (24)$$

For $H = F_{a-1}$, the lower shifts in Salikhov's notation are

$$\lambda_j = \frac{j-2}{a+1}, \quad 1 \leq j \leq a+1, \quad j \neq 2. \quad (25)$$

When a is odd, Salikhov's Theorem 1 and the parameter check in Section 4.1 show that $H, \theta H, \dots, \theta^{a-1} H$ are algebraically independent over $\mathbb{C}(z)$. When a is even, Salikhov–Viskina show that algebraic dependence of the relevant function and its first $a-1$ Euler derivatives requires one of the shifts λ_j to be an integer [9, p. 833]. None of the numbers in (25) is an integer, so the same algebraic independence holds for even a . It follows from (24) that

$$F_0, F_1, \dots, F_{a-1} \quad (26)$$

are algebraically independent over $\mathbb{C}(z)$ for every $a \geq 1$.

It remains to adjoin one function from beyond the initial range. Let

$$G := F_{a+1}, \quad \mathcal{K}_a := \mathbb{C}(z)(H, \theta H, \dots, \theta^{a-1} H).$$

The case $q = 1$, $r = 0$ of (13) is

$$\left(\frac{\theta}{a} + 1\right) G = (a + 1)! z^{-a} (F_0 - 1). \quad (27)$$

The field $\mathcal{K}_a$ is stable under differentiation by the order- a equation for H , and (26) gives $\text{trdeg}_{\mathbb{C}(z)} \mathcal{K}_a = a$.

Lemma 4.2 (Logarithmic boundary obstruction). *Every element of $\mathcal{K}_a$ has a log-free formal expansion at infinity, whereas the zero-exponential component of every formal solution G of (27) contains*

$$-a(a + 1)! z^{-a} \log z. \quad (28)$$

In particular, $G \notin \mathcal{K}_a$.

Proof. Apply Lemma 4.1 with $r = a - 1$. A formal fundamental matrix for the augmented equation of H consists of its rank-one zero-exponential solution and the a rank-one solutions with distinct exponential parts $\rho_a \zeta z$. All of these solutions are log-free. Let $\mathcal{P}_a$ be the differential field generated over $\mathbb{C}((z^{-1}))$ by the entries of this matrix inside the standard formal Picard–Vessiot extension. It is generated by Laurent series, formal powers of z , and the exponentials $e^{\rho_a \zeta z}$, without adjoining a formal logarithm. The explicit admissible-field construction in Salikhov [8, pp. 207–210] treats the logarithm as a separate transcendental generator. Hence addition, multiplication, inversion, and differentiation in $\mathcal{P}_a$ do not introduce a power of $\log z$.

The formal expansions of $H, \theta H, \dots, \theta^{a-1} H$ lie in $\mathcal{P}_a$, so the formal image of $\mathcal{K}_a$ is contained in $\mathcal{P}_a$ and is log-free. Project the right side of (27) onto its zero-exponential component. The zero component of $F_0 - 1$ has constant term -1 , and therefore the right side has leading term $-(a + 1)! z^{-a}$. The operator $\theta/a + 1$ annihilates z^{-a} . Since

$$\left(\frac{\theta}{a} + 1\right) (z^{-a} \log z) = \frac{1}{a} z^{-a},$$

the projected equation forces the term (28). A solution of the homogeneous equation is a constant multiple of z^{-a} and cannot remove the logarithm. Thus G cannot belong to the log-free field $\mathcal{K}_a$. $\square$

The lemma excludes rational membership in $\mathcal{K}_a$. The next elementary differential-field argument excludes algebraic membership.

Lemma 4.3 (Trace descent). *Let K be a differential field of characteristic zero with algebraically closed field of constants, and let $R, p \in K$. Suppose that y is algebraic over K and satisfies $Dy + py = R$. Then the equation has a solution $y_0 \in K$. If every solution of $Du + pu = 0$ in an algebraic extension of K already belongs to K , then $y \in K$.*

Proof. Put $L = K(y)$ and $d = [L : K]$. Derivation commutes with the field trace, so

$$y_0 := \frac{1}{d} \text{Tr}_{L/K}(y)$$

belongs to K and satisfies $Dy_0 + py_0 = R$. The difference $y - y_0$ satisfies the homogeneous equation. The constants of the finite algebraic differential extension L/K are algebraic over the constants of K and hence introduce no new constants because the latter field is algebraically closed. The final assertion follows. $\square$

Apply Lemma 4.3 to (27), with $D = \theta/a$ and $p = 1$. The homogeneous solutions are cz^{-a} , with $c \in \mathbb{C}$, and already belong to $\mathcal{K}_a$. If G were algebraic over $\mathcal{K}_a$, the trace lemma would therefore imply $G \in \mathcal{K}_a$, contrary to Lemma 4.2. Hence G is transcendental over $\mathcal{K}_a$, and the $a + 1$ functions

$$F_0, F_1, \dots, F_{a-1}, G \quad (29)$$

are algebraically independent over $\mathbb{C}(z)$.

These functions and the derivatives needed to close their equations form a system of E -functions over $\overline{\mathbb{Q}}(z)$ with no nonzero finite singularity. Its function field has transcendence degree $a + 1$: the derivatives of H lie in $\mathcal{K}_a$, while (27) puts every derivative of G in $\mathcal{K}_a(G)$. Beukers's Theorem 1.1 [7, Theorem 1.1] preserves this transcendence degree at the ordinary point $z = 1$. By the triangular transformation (24) and equation (27), the corresponding value field is generated by

$$F_0(1), F_1(1), \dots, F_{a-1}(1), G(1). \quad (30)$$

The values in (30) are therefore algebraically independent over $\overline{\mathbb{Q}}$. Equation (3) gives the algebraic independence of the constants in (20).

It remains to identify the rest of the fixed-slope family. Shifting the summation index gives

$$C_{a,a} = (a + 1)C_{a,0} - \frac{1}{a!} \quad (31)$$

and, for $b > a + 1$,

$$C_{a,b} = \frac{C_{a,b-1} - (a + 1)C_{a,b-a-1} + (a + 1)/b!}{b - a - 1}. \quad (32)$$

Thus every $C_{a,b}$ with $b \geq 0$ is a rational affine combination of (20). For $1 - a \leq b < 0$, set $r = a + 1 + b$. The index shift used in Section 4.1, followed by (22), gives

$$C_{a,b} = \frac{1}{r!} + \frac{a + 1 - r}{a + 1}C_{a,r} + \frac{1}{a + 1}C_{a,r-1}. \quad (33)$$

Here $2 \leq r \leq a$, and (31) eliminates $C_{a,a}$ when $r = a$. This proves the affine-span assertion in Theorem 1. The algebraic independence of (20) supplies the matching lower bound in (21) and completes the proof. In particular, the affine recurrences account for all algebraic relations at fixed slope: after any finite collection of the constants is expressed in the basis (20), its ideal of polynomial relations is generated by the resulting affine linear relations.

References

- [1] P. Erdős and R. L. Graham, *Old and New Problems and Results in Combinatorial Number Theory*, Monographies de L'Enseignement Mathématique **28** (1980), pp. 65–66.
- [2] T. F. Bloom, Erdős Problem #270, *Erdős Problems*. erdosproblems.com/270; accessed 22 August 2026.
- [3] T. Crmarić and V. Kovač, On the irrationality of certain super-polynomially decaying series, *Colloquium Mathematicum* **179** (2025), 55–68. doi:10.4064/cm9628-5-2025.
- [4] V. Kovač, comment on Erdős Problem #270, *Erdős Problems* forum, 12 July 2026. Post 7469; accessed 22 August 2026.
- [5] C. D. Long, “You can probably get algebraic independence here with a minor extension,” post on X, 23 August 2026. Online post; accessed 23 August 2026.
- [6] D. Weber and I. Pinelis, answers to “Is the integral of e^{-x^2} from 0 to 1 known to be irrational?”, MathOverflow, 26 December 2023. Online discussion.
- [7] F. Beukers, A refined version of the Siegel–Shidlovskii theorem, *Annals of Mathematics* **163** (2006), 369–379. doi:10.4007/annals.2006.163.369.
- [8] V. Kh. Salikhov, A criterion for the algebraic independence of values of a class of hypergeometric E -functions, *Mathematics of the USSR-Sbornik* **69** (1991), 203–226. doi:10.1070/SM1991v069n01ABEH002113.
- [9] V. Kh. Salikhov and G. G. Viskina, Algebraic relations between the hypergeometric E -function and its derivatives, *Matematicheskie Zametki* **71** (2002), 832–844; English translation in *Mathematical Notes* **71** (2002), 761–772. doi:10.1023/A:10158647111107.
- [10] M. van der Put and M. F. Singer, *Galois Theory of Linear Differential Equations*, Grundlehren der mathematischen Wissenschaften **328**, Springer, 2003, Chapter 3, Theorem 3.1. doi:10.1007/978-3-642-55750-7.